MESH TOPOLOGY

1. Project Overview:
2. Materials Used:

Item	Quantity	Model
Multilayered Switch	1	3560-24PS
Access Switches	5	2960-24TT
PC	10	PC-PT

3. Network design and Topology:

   Physical Topology

   - Core switch: One multilayered switch handling all inter routing
   - Access distribution: 5 access switches, with four switches connected directly to the core
   - End devices: 10 PC connected to the access switches.
4. Configuration
5. IP Addressing

   All the departments in the setup have their designated VLAN

- VLAN 10 – Admin Department
  - Network address: 192.168.10.0/24
  - Subnet Mask: 255.255.255.0
  - Default gateway: 192.168.10.1

- VLAN 20- HR Department
  - Network Address:192.168.20.0/24
  - Subnet mask:255.255.255.0
  - Default gateway: 192.168.20.1

- VLAN 30- IT Department
  - Network Address: 192.168.30.0/24
  - Subnet mask: 255.255.255.0
  - Default gateway: 192.168.30.1

- VLAN 40- Finance Department
  - Network Address: 192.168.40.0/24
  - Subnet mask: 255.255.255.0
  - Default gateway: 192.168.40.1

- VLAN 50- Legal Department
  - Network Address: 192.168.50.0/24
  - Subnet Mask: 255.255.255.0
  - Default gateway: 192.168.50.1

Device	IP address	VLAN	Remark
Admin-PC1	192.168.10.2	10	Connected to the admin switch
Admin-PC2	192.168.10.3	10	Connected to admin switch
HR-PC1	192.168.20.2	20	Connected to the HR switch
HR-PC2	192.168.20.3	20	Connected to HR switch
IT-PC1	192.168.30.2	30	Connected to IT switch
IT-PC2	192.168.30.3	30	Connected to IT switch
Finance-PC1		40	Connected to the Finance switch
Finance-PC2		40	Connected to the finance switch
Legal-PC1	192.168.50.3	50	Connected to the legal switch
Legal-PC2	192.168.50.2	50	Connected to legal switch

6. Testing and Verification

   a) The cdp neighbors as seen by the core

```
Switch#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID        Local Intrfce    Holdtme     Capability     Platform     Port ID
ITSwitch         Fas 0/3           152             S         2960         Fas 0/1
HRSwitch         Fas 0/1           152             S         2960         Fas 0/2
Admin            Fas 0/4           152             S         2960         Fas 0/1
FinanceSwitch
                 Fas 0/2           152             S         2960         Fas 0/1
```

   b) IP interfaces as seen by the core

```
Switch#
Switch#
Switch#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES unset  up                    up
FastEthernet0/2        unassigned      YES unset  up                    up
FastEthernet0/3        unassigned      YES unset  up                    up
FastEthernet0/4        unassigned      YES unset  up                    up
FastEthernet0/5        unassigned      YES unset  down                  down
FastEthernet0/6        unassigned      YES unset  down                  down
FastEthernet0/7        unassigned      YES unset  down                  down
FastEthernet0/8        unassigned      YES unset  down                  down
FastEthernet0/9        unassigned      YES unset  down                  down
FastEthernet0/10       unassigned      YES unset  down                  down
FastEthernet0/11       unassigned      YES unset  down                  down
FastEthernet0/12       unassigned      YES unset  down                  down
FastEthernet0/13       unassigned      YES unset  down                  down
FastEthernet0/14       unassigned      YES unset  down                  down
FastEthernet0/15       unassigned      YES unset  down                  down
FastEthernet0/16       unassigned      YES unset  down                  down
FastEthernet0/17       unassigned      YES unset  down                  down
FastEthernet0/18       unassigned      YES unset  down                  down
FastEthernet0/19       unassigned      YES unset  down                  down
FastEthernet0/20       unassigned      YES unset  down                  down
FastEthernet0/21       unassigned      YES unset  down                  down
FastEthernet0/22       unassigned      YES unset  down                  down
FastEthernet0/23       unassigned      YES unset  down                  down
FastEthernet0/24       unassigned      YES unset  down                  down
GigabitEthernet0/1     unassigned      YES unset  down                  down
GigabitEthernet0/2     unassigned      YES unset  down                  down
Vlan1                  unassigned      YES unset  administratively down down
Vlan10                 192.168.10.1    YES manual up                    up
Vlan20                 192.168.20.1    YES manual up                    up
Vlan30                 192.168.30.1    YES manual up                    up
Vlan40                 192.168.40.1    YES manual up                    up
Vlan50                 192.168.50.1    YES manual up                    up
```

c) The following depicts the IP route of the core switch

```
Switch#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.10.0/24 is directly connected, Vlan10
C    192.168.20.0/24 is directly connected, Vlan20
C    192.168.30.0/24 is directly connected, Vlan30
C    192.168.40.0/24 is directly connected, Vlan40
C    192.168.50.0/24 is directly connected, Vlan50

Switch#
```

d) The VLANs found on the router

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/1
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/2
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/2
                                                Gig0/1, Gig0/2
10   Admin                            active
20   HR                               active
30   IT                               active
40   Finance                          active
50   Legal                            active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

e) Pinging the gateways from the core switch

```
Switch#ping 192.168.10.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/2/8 ms

Switch#ping 192.168.20.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.20.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/3/5 ms

Switch#ping 192.168.30.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.30.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/3/6 ms

Switch#ping 192.168.40.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.40.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/2/5 ms

Switch#ping 192.168.50.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.50.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/2/5 ms
```

f) Pinging the end devices from the core switch

```
Switch#ping 192.168.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms

Switch#ping 192.168.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/4 ms

Switch#ping 192.168.30.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.30.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

Switch#ping 192.168.40.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.40.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

Switch#ping 192.168.50.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.50.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms
```

(NOTE: Before a successful reply can be received, we first got a time out because the packet was still learning the way hence the 80% success rate then 100%)

g) The cdp neighbors of the Finance department

```
FinanceSwitch>en
FinanceSwitch#show cdp neighbor
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID    Local Intrfce     Holdtme    Capability   Platform    Port ID
Switch       Fas 0/1            122                    3560        Fas 0/2
LegalSwitch  Fas 0/2            122            S       2960        Fas 0/2
```

h) The cdp neighbors of the HR department

```
HRSwitch>en
HRSwitch#show cdp neighbor
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID    Local Intrfce     Holdtme    Capability   Platform    Port ID
Switch       Fas 0/2            124                    3560        Fas 0/1
LegalSwitch  Fas 0/1            124            S       2960        Fas 0/1
```

(NOTE: it was previously mentioned that only 4 switches are connected to the core switch while one is not, the two above (g and h) is meant to show where the Legal department switch is connected.)

i)

7) Advantages
8) Disadvantages
9) Conclusion